Future-Context Requirements for Streaming Accent Conversion: A Controlled Study Using Causal Phone Translation

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***Abstract*—Streaming foreign accent conversion systems operate under a fixed lookahead budget, yet how accuracy depends on future context has not been characterised, and the budget with which a model is labelled is rarely verified against the future context that it actually uses. We study both questions with a causal phone-translation probe over a frozen self-supervised encoder. First, masking self-attention does not make WavLM or wav2vec 2.0 base causal: a positional convolution and a normalisation layer admit 1.28 s and the entire utterance of future input, respectively. We give two patches and verify them by truncation. Second, a lookahead mask applied in every layer compounds with depth: at a nominal 40 ms, the output of the 13-layer model depends on at least 220 ms of future audio. We give a perturbation test that measures effective future context and a masking scheme for which it equals the nominal value exactly. Third, across 21 per-layer budgets from 0 to 640 ms, phone error rate decreases by approximately 0.045 per doubling ($R^2 = 0.98$). Piecewise fits indicate curvature but no identifiable knee; a repeatability-aware analysis places the onset of diminishing returns near a nominal 340 ms, with a 300–480 ms sensitivity range, which corresponds to at least 1.5 s of effective context. Canonical-phone prediction benefits more from future context than realized-phone transcription, with a steeper absolute slope at canonical-deviation positions, and a pre-specified manner-class hypothesis is not supported under speaker-level resampling. A second encoder, causalised wav2vec 2.0, yields a weak, non-monotonic response (3.9% relative endpoint improvement against 63.3% for WavLM). Per-chunk compute varies only weakly with lookahead.**

***Index Terms*—accent conversion, streaming speech, latency, lookahead, self-supervised models**

I. Introduction

Streaming foreign accent conversion (FAC) has been demonstrated at low latency. PHONOS [2] operates at $\leq$40 ms lookahead and reports an 81% reduction in non-native accent confidence. Prior work does not, however, characterise how accuracy varies with the lookahead budget, what a 40 ms budget costs relative to a larger one, or how the resulting latency divides between the algorithm and the hardware.

Three gaps motivate the study.

The budget is chosen, not characterised. PHONOS *states* 40 ms and publishes no ablation showing 40 ms is necessary or sufficient; TVTSyn [3] fixes 4 future tokens ($\approx$80 ms) without varying them. DarkStream [4] does ablate, over three points $\{0, 140, 280\}$ ms, and finds the third buys $<$1pp. But three points scored on anonymisation accuracy cannot tightly localise where returns diminish, which is the question that the choice of a deployment budget requires. The benefit of an intermediate budget such as 160 ms has not been reported.

Algorithmic and computational latency are reported fused. A figure such as PHONOS's "$<$241 ms on a single GPU" [2] combines a modelling choice with unstated inference and buffering costs. It is therefore not possible to determine whether faster hardware would reduce it. We decompose throughout as

$$t_{\mathrm{e2e}} = \underbrace{c + L}_{t_{\mathrm{alg}}} + t_{\mathrm{cmp}} + t_{\mathrm{buf}}, \tag{1}$$

where c is chunk size, L is lookahead, t_{cmp} is per-chunk compute and t_{buf} is I/O and jitter buffering. The $c{+}L$ form assumes a block-emission convention: output for a chunk is produced only once the whole c ms chunk and its L ms of future context have arrived. Conventions that emit frame-synchronously, or that charge only the lookahead, give a different constant; we state ours so the terms can be re-derived under another. t_{alg} is inherent: it is the delay on an infinitely fast machine, because the model does not emit the frame ending at time t_n until it has observed input through $t_n{+}L$. In frames that is $N_L = \lceil L/h \rceil$ of them at hop h, and §III-F returns to the consequences of the rounding. t_{cmp} is implementation- and hardware-dependent and can be reduced through optimisation or faster hardware.

The nominal budget is not verified against the context used. A lookahead budget is usually imposed through an attention mask or a bounded convolution and is reported as configured. Whether the output of the complete model depends on no more future input than that budget is seldom tested. Section III-B shows that the test matters: a mask applied in every layer of a deep model yields an effective future context several times its nominal value, and Eq. 1 holds only when L denotes the former.

Degradation is reported in aggregate. Existing results are utterance-level MOS, WER or accentedness. We found no prior streaming FAC study reporting lookahead sensitivity by phonetic manner class.

Our contributions:

1) Two patches that make a masked self-supervised encoder causal, with truncation-based verification (§III-C).

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Harness, causality audit, sweep configurations, analysis scripts with self-tests and per-condition hypothesis dumps: https://github.com/Dipeshtripathi13/streaming-fac-lookahead. Data is L2-ARCTIC [1] via `KoelLabs/L2Arctic`. Given that corpus's CC-BY-NC-4.0 licence, we conservatively release configurations rather than model weights.

2) A demonstration that per-layer lookahead masks compound with depth, a perturbation test that measures the effective future context of a complete model, and a masking scheme whose effective context equals its nominal lookahead exactly (§III-B).
3) An accuracy characterisation of a causal phone-translation probe for streaming FAC across 21 per-layer lookahead budgets (a 16-point main sweep plus five targeted additions), reporting an average exchange rate and a floor-relative saturation region rather than an unidentifiable knee (§V-C).
4) A negative result on a pre-specified phoneme-class hypothesis (§V-E).
5) Evidence that canonical-phone prediction benefits more strongly from lookahead than realized-phone transcription under matched capacity, and that this holds within a single arm, with the association surviving speaker-level resampling, leave-one-speaker-out and within-phone stratification for phone identity (§V-F).
6) A cross-encoder test showing that neither the exchange rate nor the detailed curve shape transfers from WavLM base+ to wav2vec 2.0 base, replicated at a second seed (§V-D).
7) Three documented measurement failures, each of which produced plausible output before being caught (§VI).

II. Related work

Nguyen et al. [5] present a streaming non-autoregressive accent-conversion system on an Emformer encoder, demonstrating streaming feasibility; PHONOS [2] subsequently targets much lower latency with bounded future context. Neither characterises how quality varies with that context, which is the axis we isolate. PHONOS, TVTSyn [3] and DarkStream [4] originate from one group and pair conversion with speaker anonymisation. TVTSyn gives CPU as well as GPU latency ($\approx$132 and $\approx$79 ms at 60 ms chunk), so a CPU number is not unpublished; a CPU latency–*quality* curve is. DarkStream ablates look-ahead over $\{0, 140, 280\}$ ms and selects 140 ms because the extra 140 ms buys $<$1pp at double the buffering delay: a three-point ablation, scored on anonymisation accuracy rather than conversion quality. DarkStream likewise reports diminishing marginal return at its largest tested budget, though a three-point ablation on a different objective is too sparse to compare saturation locations with the estimate we obtain over 21 budgets.

StreamVC [6] reports mobile operation without device, core or thread details, limiting hardware-level reproducibility. A survey [7] frames the space; streaming anonymisation [8], [9], [10] shares the constraint but not the accent objective; offline FAC [11], [12] and low-latency VC [13] bracket the design space. Our contribution is orthogonal: we characterise an axis these systems parameterise but do not study, and claim no quality improvement over any of them.

III. Method

The study isolates one design variable, the amount of future context made available to a streaming model, and measures its effect on a supervised phonetic-normalisation task. This section defines the task and the latency model (§III-A), the masking scheme that realizes a lookahead budget (§III-B), the modifications required to make a pretrained self-supervised encoder causal (§III-C), the probe network and its matched control (§III-D), and the quantities reported in Section V (§III-E).

A. Problem formulation and latency model

Let $\mathbf{x}$ be a 16 kHz waveform from a non-native speaker and let $\mathbf{z}_{1:T}$ denote the encoder frames computed from it at hop $h = 20$ ms. L2-ARCTIC [1] annotates each utterance with two phone sequences: the canonical sequence $\mathbf{y}^{\mathrm{can}}$ derived from the lexicon (`g2p`), and the realized sequence $\mathbf{y}^{\mathrm{real}}$ transcribed from what the speaker produced (`ipa`). Training a complete FAC pipeline at every lookahead budget is computationally prohibitive, so we operationalize the phonetic-normalisation component of accent conversion as the mapping

$$f_\theta : \mathbf{x} \longmapsto \mathbf{y}^{\mathrm{can}}, \tag{2}$$

that is, the prediction of the canonical phone sequence from accented audio. This is a controlled proxy for one stage of FAC; the future-context requirements of acoustic generation and prosody may differ, and we make no claim about them.

A streaming system consumes audio in chunks of $c = Ch$ ms and may delay its output by a lookahead of L ms, realized as $N_L = \lceil L/h \rceil$ frames. The end-to-end delay then decomposes as in Eq. 1. The algorithmic term $c+L$ assumes a block-emission convention, in which the output for a chunk is produced once the chunk and its L ms of future context have arrived; conventions that emit frame-synchronously differ by a constant. The convolutional front end of the encoders studied here has a 25 ms receptive field at a 20 ms hop and therefore contributes a further 5 ms that no setting of L removes.

B. Chunked lookahead masking

Lookahead is imposed by restricting self-attention. With zero-based frame indices, chunk length C frames, lookahead N_L frames and lookback N_B frames, query frame i may attend to key frame j if and only if

$$i - N_B \leq j \leq (\lfloor i/C \rfloor + 1)\, C - 1 + N_L. \tag{3}$$

The upper limit is the last frame of the chunk containing i, extended by N_L frames: every frame of a chunk is emitted together, so frames early in a chunk may use the remainder of that chunk at no additional delay. The mask is added to the pre-softmax attention logits of every encoder layer and of every block of the probe. In WavLM it is added to the gated relative position bias, which leaves the model's own padding mask in effect. All experiments use $C = 2$ ($c = 40$ ms) and $N_B = 100$ (2 s).

Effective future context. Eq. 3 bounds the context of a *single* layer. When the same mask is applied in each of K stacked layers, the bounds compose: a frame at layer k attends to frames of layer $k-1$ that have themselves attended N_L frames ahead, so the output at frame i can depend on input as far as approximately $i + K(N_L + C - 1)$. For $N_L = 0$ the

TABLE I
EFFECTIVE FUTURE CONTEXT (MS) OF THE COMPLETE MODEL, WAVLM BASE+ TAPPED AT LAYER 9 WITH A FOUR-BLOCK PROBE, $c = 40$ MS. THE NOMINAL VALUE IS THE REMAINDER OF THE CHUNK PLUS L. PER-LAYER VALUES ARE LOWER BOUNDS AT A 10^{-4} THRESHOLD.

L (ms)	nominal	per-layer	end-to-end
0	20	20	20
20	40	$\geq$200	40
40	60	$\geq$220	60
80	100	$\geq$460	100
160	180	$\geq$780	180
340	360	$\geq$1500	360
640	660	$\geq$2820	660

bound does not grow with depth, since every attended frame lies in the same or an earlier chunk. For $N_L > 0$ it does, and L must then be read as a *per-layer* lookahead rather than as the end-to-end future context of the model.

End-to-end lookahead. The compounding is removed by confining the lookahead to a single layer. In the *end-to-end* scheme the encoder is masked frame-causally (Eq. 3 with $C = 1$ and $N_L = 0$), the first block of the probe uses Eq. 3 with the full N_L, and the remaining blocks use $N_L = 0$. An encoder frame then depends on no input beyond itself; the first block extends the context of each frame to the end of its chunk plus N_L frames; and the later blocks, which attend only to the current and earlier chunks, cannot extend it further. The future context of the complete model is therefore exactly $(C - 1 + N_L)h$, that is, the remainder of the chunk plus L. Because the encoder no longer depends on L, its features are also identical across budgets.

Measuring effective context. We verify both statements with a perturbation test on the complete model, comprising the nine encoder layers below the tap and the four probe blocks. All samples after a time τ are replaced by independent noise, and the earliest output frame whose relative change exceeds 10^{-4} is located; the interval between that frame and τ is the effective future context. The test is repeated for each alignment of τ within a chunk and the largest value is retained. Table I reports the result. Under the end-to-end scheme the measured context equals the nominal value at every budget. Under the per-layer scheme it is 220 ms at a nominal 40 ms and 1.5 s at a nominal 340 ms; these are lower bounds, since the influence of distant frames eventually falls below the threshold.

Scope of the reported results. The lookahead sweeps of Section V were obtained under the per-layer scheme, and L denotes the per-layer value throughout; Table I gives the corresponding effective context. For these conditions $c + L$ in Eq. 1 is a lower bound on the algorithmic delay rather than its value.

C. Making the encoder causal

Masking self-attention in wav2vec 2.0 or WavLM base [14], [15] does not produce a causal encoder, because two components outside the attention layers also read future input.

Positional convolution. The convolutional positional embedding is a grouped convolution with kernel size $K_p = 128$ and symmetric padding, so the embedding added to frame t is computed from frames $t - 64$ to $t + 63$, which is 1.28 s of future context at a 20 ms hop. We replace the symmetric padding with $K_p - 1$ frames of left padding, so that

$$\mathbf{p}_t = \sum_{k=0}^{K_p-1} \mathbf{W}_k \, \mathbf{z}_{t-(K_p-1)+k}, \tag{4}$$

which depends only on frames up to t. The pretrained kernel $\mathbf{W}$ is retained.

Feature-encoder normalisation. In the base checkpoints the first layer of the convolutional feature encoder is followed by a group normalisation with one group per channel, whose mean and variance are computed over the entire utterance. Every output frame therefore depends on every input sample. We replace it with a cumulative normalisation,

$$\begin{aligned} \mu_t &= \frac{1}{t}\sum_{s=1}^{t} x_s, \qquad \sigma_t^2 = \frac{1}{t}\sum_{s=1}^{t} x_s^2 - \mu_t^2, \\ y_t &= \gamma \, \frac{x_t - \mu_t}{\sqrt{\sigma_t^2 + \epsilon}} + \beta, \end{aligned} \tag{5}$$

applied per channel with the pretrained affine parameters γ and β. The large checkpoints use layer normalisation at this position and do not require the second patch.

Both patches alter the function computed by the pretrained encoder, which is then used frozen and without adaptation. The absolute error rates reported below should be read with that in mind; the comparisons across lookahead are unaffected, because every condition uses the same patched encoder.

Verification by truncation. A causal encoder F must satisfy $F(\mathbf{x})_t = F(\mathbf{x}_{1:\tau})_t$ for every frame t that ends before sample τ. We therefore compute

$$\delta = \frac{\|F(\mathbf{x})_{1:n-m} - F(\mathbf{x}_{1:\tau})_{1:n-m}\|_2}{\|F(\mathbf{x})_{1:n-m}\|_2}, \tag{6}$$

where n is the number of frames of the truncated input and $m = 8$ frames are excluded to cover the receptive field of the front end. The test uses a 4 s input truncated at 2 s with $N_L = 0$, and declares the encoder causal when $\delta < 10^{-4}$. Table II reports the four combinations of the two patches. Neither patch is sufficient alone, and applying the cumulative normalisation without the causal positional convolution increases δ: the two leakage paths interact, so removing one does not reduce truncation sensitivity monotonically. Training is aborted if this test fails.

TABLE II
TRUNCATION TEST ON `WAVLM-BASE-PLUS`: RELATIVE L_2 CHANGE δ (EQ. 6) IN EARLIER FRAMES AFTER LATER AUDIO IS DELETED.

configuration	δ	causal?
attention mask only	1.14×10^{-2}	no
+ causal positional conv	6.00×10^{-3}	no
+ cumulative normalisation	2.60×10^{-2}	no
both patches	$\mathbf{4.58 \times 10^{-6}}$	**yes**

The three unpatched values are stable across library versions and reproduce exactly. The value for the fully patched encoder lies at the level of single-precision accumulation error and varies with the `transformers` release (an earlier version

gave 6.14×10^{-6}). Both values are more than an order of magnitude below the 10^{-4} threshold, so the outcome of the test is unaffected, but the entry should be read as indistinguishable from causal at single precision rather than as a reproducible constant.

D. Probe network and matched control

The probe operates on the hidden states of one layer ℓ of the frozen encoder ($\ell = 9$ for WavLM base+; the choice for the second encoder is discussed in §V-D). It consists of four pre-norm blocks of width 768. Each block applies, with residual connections, (i) eight-head self-attention under the mask of Eq. 3; (ii) a depthwise convolution of kernel size 31 with left-only padding, followed by a pointwise convolution; and (iii) a feed-forward layer of width 2048 with GELU activation. Dropout is 0.1. A final layer normalisation and a linear layer produce posteriors over 37 symbols, the 36 phones of the corpus after stress marks are removed and the CTC blank. The probe has 24.6 M trainable parameters and is trained with the connectionist temporal classification (CTC) objective against the target sequence; hypotheses are obtained by greedy decoding. Padded positions are zeroed after every sub-layer, since the convolution and feed-forward layers do not otherwise respect the padding mask (§VI).

Matched control. Replacing the target $\mathbf{y}^{\mathrm{can}}$ in Eq. 2 with $\mathbf{y}^{\mathrm{real}}$ yields a realized-phone transcription model that shares the architecture, capacity, vocabulary, data order, seed and step count of the canonical-phone model and differs only in its target tensor. This is a closer control than a comparison between accent conversion and voice conversion systems, although the two targets still differ in label entropy, sequence length and annotation noise; the within-model analysis of §V-F addresses that residual difference.

E. Reported quantities

Phone error rate. PER is the Levenshtein distance between hypothesis and reference divided by the reference length, averaged over test utterances.

Exchange rate. The exchange rate is the slope β of the least-squares fit $\mathrm{PER}(L) = \alpha + \beta \log_2 L$ over the conditions with $L \geq 20$ ms, and is expressed in PER per doubling of lookahead.

Repeatability floor. Let σ be the standard deviation of test PER over complete repetitions of a condition at identical seed and configuration. A difference between two conditions is treated as resolved only if it exceeds 2σ.

Saturation onset. The saturation onset is the smallest sampled lookahead beyond which the PER gain per additional 20 ms frame remains below 2σ. Unlike a breakpoint, this quantity depends on the measured floor and not on the parameterisation of the lookahead axis.

Breakpoint identifiability. A one-segment and a two-segment linear fit are compared by the Bayesian information criterion [16]. A breakpoint is reported as a knee only if its location is stable under three treatments of the $L = 0$ condition on the logarithmic axis and under bootstrap resampling.

Preference margin. For a model trained on one target, the preference margin is its PER against the other target minus its PER against its own. It is a within-model difference, so effects that shift both error rates equally cancel.

Speaker-level uncertainty. Phone tokens are nested within speakers. Intervals for class-conditioned and position-conditioned quantities are therefore percentile bootstrap intervals over test speakers (2000 draws), accompanied by leave-one-speaker-out ranges, which do not rely on asymptotic arguments.

F. Quantisation of lookahead

Because lookahead is realized as $N_L = \lceil L/h \rceil$ frames, it can take only multiples of $h = 20$ ms. A transition narrower than 20 ms cannot be represented, and sampling more finely than h produces duplicate conditions that would reweight any fit. We sample every 20 ms increment up to 200 ms and a subset of the admissible budgets above it, and verify before fitting that no two conditions share the same t_{alg}.

IV. Experimental setup

Data. All experiments use the L2-ARCTIC corpus [1] in the `KoelLabs/L2Arctic` release, which contains 3599 phoneme-annotated utterances from 24 speakers of six native languages. Utterances shorter than 0.8 s or longer than 10 s are excluded. The data are divided into speaker-disjoint sets stratified by native language: one speaker per language is held out for validation and one for testing, and the remaining speakers are used for training. Test PER is computed on a fixed subset of 400 test utterances and validation PER on 200 validation utterances.

Training. The encoder is frozen and the probe is trained for 1200 steps with a batch size of 8, using AdamW (weight decay 0.01), a one-cycle learning-rate schedule with a peak of 3×10^{-4} and gradient-norm clipping at 1.0. The order in which training utterances are presented is identical across conditions. The adequacy of the fixed step budget is examined in §V-B.

Sweeps. Two sweeps were run on an NVIDIA T4 under the per-layer scheme of §III-B. The first covers seven lookahead budgets, both targets and three seeds (42 runs). The second is a dense sweep of 16 budgets and two seeds on both targets (64 runs). The canonical-phone arm of the dense sweep was subsequently extended with five budgets (220, 260, 300, 340 and 360 ms) to resolve the region of diminishing returns, giving 21 budgets in total.

Compute measurements. Per-chunk compute was measured on an Apple M4 (ten CPU cores, four performance and six efficiency; 16 GB; macOS 25.5) and on an ARM64 Linux host (aarch64, four logical cores, 3.8 GB, Linux 6.8, scipy-openblas 0.3.29; the platform does not expose a CPU model string). Timings use CUDA events or `perf_counter_ns`, with 28 to 50 repetitions after three discarded warm-up iterations.

Repeatability floor. A conventional multi-seed design does not isolate run-to-run variation at a fixed seed: pairing within seed cancels seed effects but leaves this residual, which is consistent with nondeterministic GPU kernels. We

measured it directly by repeating four conditions ($L \in \{40, 160, 240, 320\}$ ms) four times each at identical seed and configuration, which gives 12 degrees of freedom on test PER:

$$\sigma = 0.00146, \qquad 2\sigma = \mathbf{0.0029}\ \text{PER}, \tag{7}$$

with a bootstrap 95% interval of $[0.0013, 0.0034]$ on 2σ.

This value supersedes an earlier estimate of $2\sigma = 0.0044$ obtained from six unplanned repetitions, which the present interval excludes; the earlier estimate was inflated by a single outlying pair. The correction is material because the saturation onset of §V-C is defined relative to this floor, and a smaller floor makes smaller gains resolvable. The floor is also not constant across conditions: the per-condition σ ranges from 0.00064 to 0.00223, a factor of 3.5. A single global threshold is therefore an approximation, and decisions close to it should be interpreted accordingly.

V. Results

A. Per-chunk compute depends only weakly on lookahead

Across $L \in [0, 640]$ ms at chunk 40 ms on the M4, the nominal algorithmic term $c + L$ spans 40–680 ms (17×), and the effective future context of Table I spans a wider range still, while median per-chunk compute moves from 43.23 ms to at most 44.06 ms, a range of -0.3% to $+1.9\%$. On ARM64 the span is 0 to $+7.6\%$. The two terms of Eq. 1 are therefore separately actionable, which is the practical justification for reporting them separately.

Separability does not imply feasibility. At this configuration $t_{\text{cmp}} \approx$ 43–44 ms against a 40 ms chunk, i.e. RTF $= t_{\text{cmp}}/c \approx 1.08 > 1$: the base encoder does not sustain real time at 40 ms chunks on this machine, whatever L is. Eq. 1 describes the delay of a single chunk and must be read alongside the steady-state throughput constraint

$$\text{RTF} = t_{\text{cmp}}/c < 1, \tag{8}$$

which lookahead leaves untouched but chunk size and hardware do not. Increasing L adds little per-chunk compute over the tested range; being able to run at all is a separate constraint.

B. Is the curve an optimisation-rate curve?

Every condition elsewhere is trained for a fixed 1200 steps, so what is measured directly is *lookahead → performance after 1200 optimisation steps*. That equals *lookahead → attainable performance* only if the conditions sit at comparable points in optimisation. If longer lookahead simply trained faster, part of the exchange rate would be an optimisation-rate difference rather than a context effect. We test this possibility directly.

Five representative budgets ($L \in \{0, 40, 160, 240, 640\}$ ms) were trained for 6000 steps, five times the standard budget, with validation every 500 steps, giving a 12-point trace per condition (Table III).

Three observations follow. **Training largely flattens**: over the final 1500 steps validation PER moves by only 0.0005–0.0045 across the five conditions, the two extremes (L=0 at 0.0041, L=640 at 0.0045) still drifting slightly more than the middle three, so 6000 steps is near but not fully at convergence. We deliberately do not test these against the repeatability floor of Eq. 7: that floor estimates fixed-seed variation of *test* PER under complete reruns, which is not the same object as a validation checkpoint trajectory within a single run. **The ranking never reorders**: the ordering L=0 > 40 > 160 > 240 > 640 holds at **12 of 12** checkpoints, including the earliest at step 500. **The exchange rate is stable**: fitted over $L \geq 40$ it stays within -0.0396 to -0.0461 PER per doubling at every checkpoint. Validation runs every 500 steps, so this run has no checkpoint at the 1200-step budget used elsewhere; the nearest is step 1000, whose fitted slope (-0.0405) agrees with the 6000-step value (-0.0414) to within 0.0009 PER per doubling. The bracketing checkpoints agree too (-0.0461 at 500, -0.0397 at 1500), so the budget used elsewhere sits in a flat region of this curve rather than at a favourable point on it.

Improvement from step 1500 to 6000 is also roughly uniform across conditions (-0.030 to -0.045 PER) rather than concentrated at long lookahead, which is the specific pattern a convergence-rate confound would produce. Absolute PER does keep falling with longer training, so the fixed-budget numbers elsewhere are a comparison at matched budget rather than converged performance. But the *relative* quantities this paper reports, the ranking and the exchange rate, are not artefacts of stopping at 1200 steps. The check is single-seed over five budgets and uses validation rather than test PER, so it bounds the concern rather than eliminating it.

TABLE III
Convergence check: validation PER against optimisation step, single seed. The ranking is preserved at all 12 checkpoints and the fitted exchange rate is stable from step 1000 onward.

step	L=0	40	160	240	640
500	0.5301	0.4375	0.3288	0.2946	0.2574
1000	0.4715	0.3601	0.2588	0.2353	0.2012
2000	0.4480	0.3392	0.2332	0.2054	0.1717
3000	0.4340	0.3293	0.2131	0.1930	0.1622
4000	0.4135	0.3152	0.1992	0.1873	0.1533
5000	0.4092	0.3108	0.1962	0.1838	0.1473
6000	0.4060	0.3100	0.1951	0.1822	0.1468

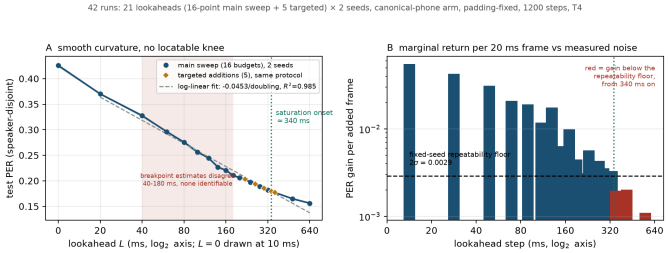


Fig. 1. Dense sweep, canonical-phone arm, 2 seeds: the 16-point main sweep (circles) with the five targeted saturation-region additions (diamonds). **Left:** smooth curvature; the shaded band spans the breakpoint estimates produced by three defensible treatments of L=0, none identifiable. **Right:** marginal return per added 20 ms frame against the measured 2σ floor; red bars are below it.

C. RQ1: an exchange rate and a saturation onset, not a knee

Canonical-phone-arm PER falls monotonically from 0.4256 at $L{=}0$ to 0.1561 at $L{=}640$ ms. Fitting $L \geq 20$ ms on $\log_2 L$ gives

$$\Delta\mathrm{PER} = -0.0452 \text{ per doubling}, \quad R^2 = 0.983, \qquad (9)$$

over the 15 positive-lookahead points of the main sweep. The five budgets added later to resolve the saturation region were sampled where the curve is flattest, so we report the fit with and without them rather than silently choosing one: including all 20 positive-lookahead conditions gives -0.0453 per doubling at $R^2 = 0.985$. Targeted sampling did not move the exchange rate.

There is curvature, and it is not localisable. BIC prefers a two-segment fit under all three axis treatments we tried ($\Delta\mathrm{BIC} = -31.0$, -40.0, -46.5 over all 21 budgets), which by the usual thresholds [16] is strong evidence. But the breakpoint lands at 40, 180 and 180 ms respectively, and the bootstrap 90% intervals span 3.2, 2.6 and 3.2 octaves ($[40, 360]$, $[60, 360]$, $[40, 360]$ ms). Twenty-one budgets do not yield an identifiable knee: the five budgets added in the region where a knee would have to lie sharpened the evidence for curvature without moving the breakpoint. The cause is structural: on a $\log_2(L{+}1)$ axis the $L{=}0 \to 20$ ms gap is 4.39 units while every other gap is $\approx$1.0, so a piecewise fit can be strongly biased toward placing a breakpoint just past that gap even when the underlying curve has no localised knee.

What we report instead is the lookahead beyond which marginal return falls below the measured floor. Because the original grid was 20 ms-spaced only to 200 ms, we added the five budgets 220, 260, 300, 340 and 360 ms under identical settings, making the range 200–360 ms fully 20 ms-spaced. The quantity below is therefore a per-*observed*-step gain, not an average over wider sampled intervals.

Gains run $+0.056$ (0→20 ms), $+0.021$ (60→80) and $+0.0045$ (180→200). Against the measured floor of 0.0029 (Eq. 7), every observed 20 ms step remains above it out to 340 ms, at $1.1\times$ (200→220), 2.0, 1.3, 1.5, 1.2, 1.1 and 1.1. The observed $340 \to 360$ ms increment is the first to fall below it, at $0.7\times$, and the average per-20 ms gain over the two longer intervals that follow stays below it as well (0.7 over $360 \to 480$ and 0.4 over $480 \to 640$). We therefore place **saturation onset at $\approx$340 ms**. The wording is deliberate: above 360 ms the grid is not 20 ms spaced, so a single sub-interval could in principle exceed the floor without lifting its interval average, and what we can state is the first sustained transition rather than a proven upper bound.

This figure moves with the floor, and we report the sensitivity rather than a single number. Recomputing the onset at each end of the floor's bootstrap interval moves it from 300 ms (at $2\sigma = 0.0034$) to 480 ms (at 0.0013). This span is a sensitivity range induced by uncertainty in the floor, not a confidence interval on a fitted saturation parameter. Under the superseded 0.0044 estimate it would have been 240 ms, so the earlier figure was an artefact of an over-conservative floor, not of the curve. The gains between 200 and 340 ms sit at only 1.1–2.0$\times$ the floor, so this is a region where returns become marginal rather than a sharp cutoff. Unlike a knee, though, the quantity depends on the floor rather than on an axis choice, and the floor is now measured with 12 degrees of freedom.

TABLE IV
CANONICAL-PHONE-ARM PER AGAINST LOOKAHEAD (MEAN OF 2 SEEDS) AND MARGINAL RETURN PER ADDED 20 MS FRAME, IN UNITS OF THE MEASURED $2\sigma{=}0.0029$ FLOOR (EQ. 7). BELOW 1.0×, THE GAIN FROM AN ADDITIONAL FRAME IS SMALLER THAN THE MEASURED FIXED-SEED REPEATABILITY THRESHOLD. THE GRID IS 20 MS-SPACED TO 360 MS; THE FINAL TWO ROWS AVERAGE OVER WIDER INTERVALS.

L (ms)	PER	gain/frame	× floor
0	0.4256	–	–
20	0.3700	0.0556	19.2
40	0.3275	0.0426	14.7
60	0.2963	0.0312	10.8
80	0.2754	0.0209	7.2
100	0.2564	0.0190	6.6
120	0.2446	0.0118	4.1
140	0.2269	0.0176	6.1
160	0.2206	0.0064	2.2
180	0.2106	0.0099	3.4
200	0.2062	0.0045	1.5
220	0.2030	0.0031	1.1
240	0.1973	0.0057	2.0
260	0.1934	0.0039	1.3
280	0.1891	0.0043	1.5
300	0.1855	0.0036	1.2
320	0.1823	0.0032	1.1
340	0.1790	0.0033	1.1
360	0.1770	0.0020	**0.7**
480	0.1649	0.0020	**0.7**
640	0.1561	0.0011	**0.4**

Gain available beyond a small budget. PER at $L{=}40$ ms is 0.3275; at 340 ms it is 0.1790, a further relative reduction of 45.3%, and at 640 ms the reduction is 52.3%. In absolute terms, of the 0.2466 PER separating $L{=}0$ from 340 ms, the 40 ms condition has captured 0.0981, or 39.8%. These budgets are per-layer values. By Table I the 40 ms condition already draws on at least 220 ms of future audio and the 340 ms condition on at least 1.5 s. The comparison therefore shows that, for WavLM base+, the probe continues to benefit from future context well beyond 200 ms. It does not measure what a system restricted to 40 ms of end-to-end lookahead forgoes, and no comparison with the budget of a published system is implied. §V-D further exhibits a backbone on which additional context yields almost no improvement.

D. The exchange rate does not transfer across encoders

Every number above is measured over one frozen backbone, which raises the question of whether the exchange rate is a property of the task or of the encoder. We repeated the sweep on a second causalised encoder, wav2vec 2.0 base [15], holding the corpus, split, head, optimiser, step budget, seed, chunk, lookback and matched lookahead grid fixed. The pretrained backbone and the selected phonetic tap layer both differ, for reasons set out below. The matched grid is $L \in \{0, 20, 40, 80, 160, 320, 640\}$ ms, a subset of the main sweep; fitted over this reduced grid the WavLM slope is -0.0446 rather than the -0.0452 of Eq. 9, which is why Table V differs slightly from §V-C. Both encoders are causalised by the same two patches and both pass the same truncation-based causality test.

TABLE V
SAME SWEEP, SECOND ENCODER, EACH TAPPED AT ITS SELECTED PHONETIC LAYER, GIVEN IN PARENTHESES (§V-D). SLOPE IS PER PER DOUBLING OF LOOKAHEAD, FITTED OVER THE MATCHED GRID. THE TWO START WITH SIMILAR PER AT L=0 BUT EXHIBIT MARKEDLY DIFFERENT LOOKAHEAD RESPONSES.

encoder	L=0	L=640	rel. gain	slope
WavLM base+ (9)	0.4256	0.1561	+63.3%	−0.0446
wav2vec 2.0 base (2)	0.4359	0.4189	+3.9%	−0.0015
seed 7	0.4376	0.4265	+2.5%	–

The layer must be matched by function, not by index. Tapping wav2vec 2.0 at layer 9, WavLM's phonetic layer, produces a degenerate run: PER rises from 0.772 at L=0 to 0.928 and the training loss is still falling steeply at step 1200. Probing the stack at L=160 ms with everything else fixed shows why (PER by layer: 0.402 at 2, 0.411 at 3, 0.437 at 4, 0.586 at 6, 0.880 at 8, 0.928 at 9, 0.971 at 10). Decodability falls monotonically with depth: this encoder carries phones low in the stack, and layer 9 is seven layers past its usable representation. Matching two encoders by layer index is not matching them.

How each layer was selected, and whether the choice drives the result. The two procedures are not symmetric and we state them plainly. WavLM's layer 9 was fixed in advance from prior practice, before any lookahead data existed, and was never re-selected. wav2vec 2.0's layer was chosen by the probe above, which runs at L=160 ms, inside the sweep it is later used to interpret, so layer and lookahead sensitivity rest on partly overlapping evidence. The concern this raises is that a layer picked mid-curve might flatter or flatten the curve it is picked from. It does not here: re-selecting on PER at L=0 alone, which is the endpoint the relative gain is measured *from* rather than a point on the response, returns the same layer (0.4359 at 2, 0.4590 at 4, 0.5735 at 6, 0.6940 at 8). The comparison below is therefore insensitive to which of the two criteria is used, though it would be cleaner still to fix both layers by a pre-specified rule.

Direction replicates; magnitude does not. At the selected layer 2, wav2vec 2.0 starts within 0.011 PER of WavLM (0.4359 against 0.4256) and then barely moves: 3.9% relative gain against 63.3%, and −0.0015 PER per doubling against −0.0446. The log-linear fit that gives $R^2 = 0.989$ on WavLM gives $R^2 = 0.099$ here, and the curve bottoms out at L=160 ms before rising again. The endpoint gain nonetheless exceeds the fixed-seed repeatability floor and is reproduced at a second training seed.

A second seed reproduces the weak, non-monotonic response. The comparison carries a headline claim, so we replicated it at a second training seed on five budgets, including the 340 ms onset that the first grid did not sample. Seed 7 lands within 0.008 PER of seed 1337 at every shared budget (0.4376/0.4359 at L=0, 0.4219/0.4169 at 40, 0.4043/0.4004 at 160, 0.4265/0.4189 at 640; seed 7 gives 0.4155 at 340, where seed 1337 sampled 320 and gave 0.4028). Both curves reach their minimum at the same L=160 ms and both rise again by 640 ms, so the non-monotonicity is a reproducible feature of this encoder rather than one seed's noise. The endpoint gain is +2.5% against +3.9%: the effect is small in both seeds, with the same non-monotonic pattern.

Nor is it an artefact of the training budget. A second seed cannot rule out an optimisation effect the two seeds share: perhaps wav2vec 2.0's lookahead conditions simply separate later. We therefore give this encoder the same convergence check WavLM receives in §V-B, at $L \in \{0, 160, 640\}$ ms to 6000 steps. Every condition improves (0.4359 → 0.3782, 0.4004 → 0.3361, 0.4189 → 0.3714), but they improve together, and the three-point ordering is unchanged: the minimum remains at L=160 ms and L=640 remains worse than it. Three budgets test that ordering, not the full seven-point curve. The endpoint gain does not grow with training; it *shrinks*, from +3.9% to +1.8%, while WavLM at the same 6000 steps gains +65.8% on test PER (0.4135 → 0.1414). Both encoder figures are test PER and so are comparable to each other; Table III reports *validation* PER at checkpoints, on which the corresponding WavLM gain is +63.8%. Five times the budget widens the gap rather than closing it.

This is not an artefact of tapping a shallow layer. Layer 2 has had substantially less contextual mixing than deeper layers, so the natural objection is that there was less context to withhold and that sensitivity would recover deeper in the stack. We do not observe that recovery. Relative gain from L=0 to 640 ms is +3.9% at layer 2, +5.4% at 4, +5.2% at 6 and −14.1% at 8: flat across every layer where the encoder can still decode phones, then negative. The expected trade-off is not observed.

Tap depth and effective context. Under the per-layer scheme the two encoders do not receive the same future context at the same L: the WavLM features pass through nine masked encoder layers and the wav2vec 2.0 features through two, each followed by the four probe blocks. Part of the difference between the two responses may therefore reflect effective context rather than the encoder. The depth check above argues against this as the whole explanation, since tapping wav2vec 2.0 at layers 4 and 6 increases the effective context without restoring the sensitivity, but decodability also falls with depth, so that check does not isolate the effect. The comparison is between two configurations that differ in both encoder and effective context.

Two encoders with similar zero-lookahead PER therefore differ by more than an order of magnitude in endpoint relative gain, a statement that holds for both wav2vec 2.0 seeds (+2.5% and +3.9% against +63.3%). **The measured −0.045 PER per doubling is specific to the WavLM base+ configuration tested here, and should be read as one encoder's operating curve rather than as a fact about how much future context English phones require.** The swap rules out treating the measured WavLM exchange rate as a backbone-independent law, and it also shows that the detailed shape need not carry over: WavLM's smooth diminishing-return curve has no counterpart in wav2vec 2.0's weak, non-monotonic one. We do not run the breakpoint and axis-perturbation analysis of §V-C on a curve that is not monotone, so we make no knee claim about it either way. Every positive quantity here, the exchange rate, the saturation onset and the gain available

beyond a small budget, should therefore be re-measured on whatever backbone a system actually deploys. Whether the gap traces to pretraining scale (960 h against 94 k h), to WavLM's denoising and speaker-conditioned objectives, or to something else is not something a two-encoder comparison can decide.

E. RQ2: the pre-specified manner grouping does not predict lookahead benefit

We fixed this prediction in code before the hypothesis outputs it is tested on were generated. The grouping, and the rationale for it, sit in `eval/phoneme_analysis.py` at commit `c5066dc` of the public repository (2 Aug 2026),[1] seven days before the hypothesis dumps were committed (`b66d0a5`, 9 Aug 2026). We are precise about what this does and does not establish. It is not a registered pre-registration, and we do not call it one: git author and committer dates are set by the committing machine and history can be rewritten, so the chronology is a claim about our repository rather than third-party attestation. What it does give a reader is the *content* of the prediction, verbatim and inspectable, including the class assignment and the reasoning we offered for it at the time. The current public history is consistent with the prediction preceding the outputs it is tested on, but consistency is the most that can be claimed: this is not immutable third-party timestamping. The prediction was that vowels and approximants break first while stops, fricatives, affricates and nasals would survive short lookahead. The reasoning was that vocalic identity is carried by formant movement across a syllable nucleus that is tens to hundreds of milliseconds long. Peterson and Lehiste measured English syllable nuclei at roughly 80–250 ms [17], a range consistent with the vowel durations later reported over 139 talkers [18]. The second half of the reasoning was that this movement is criterial rather than incidental, since even nominal monophthongs are identified better from a trajectory than from a steady-state slice [19], with glides and diphthongs differing from vowels in rate of formant change rather than in kind [20]. Stops, fricatives and affricates were placed in the comparison group because their defining cues are comparatively localised: stop place is signalled by brief formant transitions to a locus [21] and is recoverable from a short-time spectrum at the release [22], so a narrow window should suffice. Nasals were assigned to the same group under the same short-context prediction, but the acoustic argument above is strongest for the obstruent classes and we do not claim it extends cleanly to nasals. We record this asymmetry as it stood when it was specified rather than reconstructing a stronger rationale after the grouping failed. Nasals in fact gain the most of any class (Table VI). Testing this on 42,255 unique reference-phone tokens evaluated across seven lookahead conditions (295,785 phone–condition observations), with each substitution or deletion charged to the class of the *reference* phone:

Group means differ in the predicted direction (0.676 vs 0.636), but that is the weakest available reading and four stronger checks fail. The effect size is 0.66: the between-group difference (0.040) is smaller than the pooled SD across classes (0.060). The surviving group's *internal* spread is 0.190, which is 4.7× the between-group difference. And 5 of 12 pairwise orderings are violated, with nasals out-gaining all three breaks-first classes.

TABLE VI
RELATIVE ERROR-RATE REDUCTION, $L{=}0 \rightarrow 640$ MS. B = PRE-SPECIFIED AS BREAKING FIRST, S = AS SURVIVING. INTERVALS ARE PERCENTILE BOOTSTRAP OVER THE SIX TEST SPEAKERS, NOT OVER PHONE TOKENS. THE PREDICTED GROUPING DOES NOT ORDER THE DATA.

	class	rel. gain	95% CI	ref. phones
S	nasal	0.716	[0.668, 0.779]	4 020
B	approximant	0.700	[0.628, 0.774]	4 575
S	fricative	0.687	[0.608, 0.774]	7 659
B	vowel (mono)	0.673	[0.595, 0.767]	13 008
B	vowel (diph)	0.654	[0.577, 0.758]	4 824
S	stop	0.615	[0.550, 0.689]	7 071
S	affricate	0.525	[0.397, 0.633]	1 098

Under speaker-level resampling the difference is not distinguishable from zero. The 42,255 tokens per condition are not independent: they are nested in six speakers, and the claim at issue is about accented speech, for which the independent replicate is a talker rather than a phone. Resampling speakers rather than tokens (2000 draws, Table VI) puts the between-group difference at $+0.040$ with a 95% interval of $[-0.010, +0.101]$, which **includes zero**. Every individual class gain stays substantial, with the intervals in Table VI between 0.40 and 0.78; what fails to separate from zero is the *grouping*. Token-level intervals would have been roughly an order of magnitude narrower and would have declared this difference significant, which is the error the clustering avoids. The same procedure applied to RQ3 (Sec. V-F) resolves the larger interaction there, so the estimator is not uniformly uninformative at six speakers, though that alone does not establish power for an effect this small. A cluster bootstrap over six clusters leans on asymptotics that six clusters do not supply, so we add the non-asymptotic companion: recomputing the difference with each speaker removed in turn gives $+0.0217$ to $+0.0592$ (pooled $+0.0401$). No single talker drives the estimate, since the sign is stable across all six drops, but the entire leave-one-out range also sits inside an interval that contains zero, so the verdict does not change.

The pattern does not hold within L1 either, but that split cannot carry weight. Splitting by native language, the ratio of group gains straddles unity: Vietnamese 1.23, Korean 1.19, Arabic 1.08, Spanish 1.02, Hindi 0.99, Chinese 0.92; requiring both the gain ratio and the mean slope to move in the predicted direction, only 3 of 6 qualify (Spanish clears the gain criterion but not the slope one). The reversals show that the pooled ordering is not consistent across the six held-out speakers, each of whom carries a distinct L1 label. The deeper limitation is that in the speaker-disjoint test split *each L1 is represented by exactly one speaker* (ABA, LXC, SVBI, HJK, EBVS, PNV), so L1 and talker are perfectly confounded: this table cannot separate a language effect from an individual one, and every "by-L1" row is equally readable as one speaker's idiosyncrasy. We report it as a consistency check, not as evidence about native language.

[1] https://github.com/Dipeshtripathi13/streaming-fac-lookahead/commit/c5066dc

H2 is not supported. Not because lookahead fails to help vowels, since every class improves 52–72% relative, but because the pre-specified manner grouping does not reliably order lookahead benefit.

Reweighting each error by a proxy for its acoustic cost does not rescue it. PER charges every error exactly 1.0, so the natural objection is that the grouping is sound and PER simply mis-weights it: a substituted vowel and a substituted stop cannot sound equally wrong. We test that directly.

Each utterance's canonical phone string is synthesised with a single fixed voice and force-aligned to its own synthesis using an espeak-IPA CTC model, giving a time span per reference phone. The hypothesis string is synthesised with the same voice and DTW-aligned to the reference, and the spectral distortion inside each reference phone's own span is charged to that phone's class. Localisation matters here: a VITS-class synthesiser is globally coupled, so changing one phone perturbs the whole utterance and a whole-utterance distortion cannot be attributed to a single phone. The per-phone span from forced alignment is what makes the measurement local at all.

Synthesis must also be deterministic. Piper defaults to $\mathrm{noise_scale} = 0.667$ and $\mathrm{noise_w} = 0.8$, under which the same phone string renders differently on each call: self-distortion is 244.8 against 277.6 for a genuine one-phone change, so roughly 88% of an apparent "cost of an error" would be synthesis noise. Both terms are zeroed, giving bitwise-identical renderings, and the analysis refuses to run otherwise.

Two sanity checks support the measure before it is used, and the verdict is suppressed if validation fails: phones the sequence metric scores as incorrect carry $1.72\times$ the distortion of correct ones (86.8 versus 50.5), and mean distortion falls monotonically with lookahead (71.4 to 46.9, -34%).

The verdict is unchanged. Acoustic weighting does move in H2's favour, with effect size $0.607 \to 0.756$ and ordering violations $4/12 \to 2/12$ against the sequence metric on the same utterances, but not far enough to clear the pre-specified threshold of 1.0, and the failure has the same shape as before: the "survives" group's internal spread (0.120) is $3.5\times$ the difference between the groups (0.034), and a nasal again outgains a diphthong; the per-class figures are in the repository. So this particular equal-cost objection is not supported by the synthesised distortion analysis; it does not exclude weightings this proxy cannot see, notably trajectory errors at correctly identified phones.

One explanation does remain open, and it is not the one this test closes. Both signals here are TTS renderings of phone strings, so the synthesiser produces canonical formant trajectories for whatever string it is given. This measures the cost of getting a phone's *identity* wrong; it says nothing about a wrong trajectory for a phone whose identity is right. That still needs real converted audio against a native rendition (§VII). The sample is also far smaller than the sequence-level test, at 1451 unique reference phones per acoustic condition against 42,255 per sequence condition, and is weighted accordingly.

F. RQ3: canonical-phone prediction benefits more from lookahead than realized-phone transcription

Identical architecture, capacity, data order, seed and step count; the two arms differ only in the target tensor. Endpoint gain $L{=}0 \to 640\,\mathrm{ms}$ is -0.2695 (63.2% relative) for canonical-phone prediction against -0.1893 (42.8%) for realized-phone transcription, a paired difference of $+0.0802$ across the three seeds ($\mathrm{sd} = 0.0035$, $t(2) = 39.7$, $p = 6.3 \times 10^{-4}$, 95% CI $[+0.0715, +0.0889]$). With three seeds the interval is wide in principle and narrow here only because the seed-to-seed spread is small relative to the effect.

The dense grid replicates this, and sharpens it. Running the full 16-lookahead $\times$ 2-seed sweep on the transcription arm as well puts both arms on the same dense grid. The endpoint gains reproduce the 7-point result almost exactly, at 63.3% for canonical-phone prediction against 43.5% for transcription, against 63.2% and 42.8% before, in an independent set of training runs on a denser grid, though on the same corpus and test split; per-condition values for both arms are in the repository. The exchange rates separate cleanly:

$$\text{canonical-phone } -0.0452 \text{ vs transcription } -0.0292 \text{ PER per doubling,}$$

$R^2 = 0.983$ and 0.984 respectively, so **canonical-phone prediction buys $\mathbf{1.55}\times$ as much per doubling of lookahead**.

The effect is clearer in the between-arm gap than in the absolute levels. At $L{=}0$ the two arms have similar PER (0.4256 vs 0.4427, a gap of 0.0171); by $L{=}640$ ms the gap is 0.0939, **5.5$\times$ wider**. It widens in 13 of 15 adjacent steps, and the transcription arm is worse at 16 of 16 lookaheads, every one by more than the 0.0029 floor.

The arms differ in their target tensor, which carries more than the task: label entropy, sequence length and annotation noise move with it. We therefore repeat the test within a single canonical-phone model, holding the target representation and the model fixed; the two position subsets can of course still differ in composition and baseline difficulty, which is what the controls below address.

Defining a canonical-deviation position. The `g2p` and `ipa` strings are not positionally aligned and differ by substitutions, deletions and insertions, so a position-i comparison is not available. We align them with Levenshtein backtrace under unit substitution, insertion and deletion costs, and index every quantity by the *canonical* (`g2p`) sequence. A position is a canonical-deviation position when its aligned operation is a substitution or a deletion; an insertion in the `ipa` stream consumes no canonical position and is skipped. Model errors are charged to canonical positions by the same rule, aligning `g2p` against the CTC hypothesis, so that deviation status and error status index the same sequence. Where ties occur the backtrace prefers substitution over deletion over insertion, making the mapping deterministic. One consequence belongs with the result: because everything is indexed to canonical positions, realized-phone insertions are excluded from this analysis, so it covers the substitution and deletion deviations rather than every way the two strings can differ. This is the convention used throughout the paper, including the manner-class analysis of §V-E.

Splitting canonical positions this way, the exchange rate is -0.0520 PER per doubling at canonical-deviation positions against -0.0394 at matching ones, a factor of $\mathbf{1.32}$ from the same model on the same utterances. That converges with the $1.55\times$ between arms. One caveat belongs with it: in *relative* terms the ordering reverses (51% against 71%), because canonical-deviation positions start far harder (0.590 against 0.354) and the gap between the two widens with lookahead. Absolute PER decreases faster at canonical-deviation positions, although these positions begin substantially harder and show a smaller relative reduction.

The interaction survives two controls. Those positions are not a random subset: they differ in baseline difficulty, phone composition and distribution over speakers. Treating 42,255 phone tokens nested in six speakers as independent would substantially overstate precision by ignoring within-speaker dependence, so we bootstrap over *speakers*: the slope difference is -0.0127 with a 95% CI of $[-0.0211, -0.0069]$, excluding zero, and deviation positions are steeper in 2000 of 2000 resamples. Leave-one-speaker-out agrees and is not asymptotic: dropping each talker in turn gives -0.0152 to -0.0095, negative in all six cases and never approaching zero. The result is also insensitive to how ambiguous alignments are resolved: re-running with the backtrace preferring deletion over substitution moves the slope difference from -0.0127 to -0.0131 (ratio 1.32 to 1.33, CI $[-0.0218, -0.0071]$, still 26 of 35 phones in the same direction), so the result does not depend on the tie-breaking rule. Recomputing the difference *within each reference phone* and pooling gives -0.0098, steeper in 26 of 35 phones, so phone composition accounts for roughly a quarter of the raw gap but not the effect. We report this as an association under matched conditions, not as evidence that `g2p≠ipa` positions are exclusively where accent resides: such positions also include pronunciation variants, reductions and annotation error.

One secondary observation follows: neither arm yields a locatable knee (breakpoints move to $[40, 240, 160]$ ms across axis treatments for transcription, against $[40, 180, 180]$ for canonical-phone prediction), so §V-C's structural conclusion is not an artefact of the canonical-phone arm. We do not compare saturation onsets between the arms: the five budgets added over 200–360 ms to locate it were run on the canonical-phone arm only, so the transcription arm lacks the resolution the comparison would need. The transcription arm's own curve is just as log-linear, with a shallower slope.

A complementary within-run measure is the *preference margin*, PER against the other arm's labels minus PER against its own. It grows monotonically in 3/3 seeds for the canonical-phone arm ($+0.032$ at $L{=}0$ to $+0.099$ at 640 ms; 18/18 adjacent step $\times$ seed comparisons positive) while the transcription control is 0/3 monotone and drifts *down* (-0.0103). The margin is a difference rather than a level, which makes it robust to the offset discussed in §VI.

G. t_{buf}: jitter measured, I/O not

t_{buf} splits into a jitter term and an I/O term. The first is measured; the second resisted three separate attempts, and we report that rather than substitute a number.

On the M4 test platform, measured jitter is negligible relative to the other latency terms. Two independent 30 s duplex runs on the M4 at 40 ms blocks give required jitter buffers of 0.14 and 0.08 ms, p99 callback lateness $\approx$0.06 ms, zero over- or underruns, and clock drift below $1.3 \times 10^{-4}\%$. On this platform the jitter safety margin contributes nothing meaningful to the budget.

The I/O term is unmeasured, and the reason is instructive. Three attempts failed, each for a different reason, and we report that rather than substitute a number. An acoustic loopback was defeated by platform echo cancellation: all 20 repetitions fell below the correlation threshold, and captured level was *invariant* to output amplitude, so a louder probe could not help. The driver-reported figure is not a substitute, being identically 201.69 ms in excess of the requested block at both 10 and 20 ms; a quantity invariant to a doubling of the block size is not measuring the block. A digital loopback through a virtual audio device returns a signal so weak that the detection threshold has to be lowered until repetitions pass, which yields p50 206 ms against p90 491 ms. We reject that as threshold-dependent rather than representative, and such a device would in any case measure its own buffering. Closing this term needs a wired loopback through real converters, or a platform without echo cancellation in the path; the full traces are in the repository.

VI. Measurement audits and failure modes

A padding error changed the shape of the curve. A masked block handed a zero-padded hidden-state tensor to a depthwise convolution and feed-forward that do not respect the key-padding mask. The correction was not a constant offset: it grows with lookahead, from -0.014 PER at $L{=}0$ to -0.086 at 320 ms, $16.4\times$ the noise floor. The buggy curve therefore *understated* the benefit of lookahead, biasing the central quantity of RQ1. A constant offset would have cancelled in every within-run comparison; this did not. Every number reported in this paper was regenerated after the fix.

A forced one-breakpoint estimator returns a breakpoint even when no localised knee exists. ΔBIC cannot distinguish "there is a knee" from "piecewise fits a curve better than a line". Identifiability has to be tested by perturbing the axis and bootstrapping the location, not inferred from ΔBIC alone. Our synthetic check reproduces the artefact exactly: on a curve that is log-linear for $L \geq 20$ with $L{=}0$ off the extrapolation, including $L{=}0$ yields $\Delta\mathrm{BIC} = -39$ and a confident "knee at 40 ms", while dropping it yields $+5$ and no knee at all. An earlier maximum-distance-to-chord estimator reported "knee at 160 ms, bootstrap 95% CI $[160, 160]$" on this data, a tight interval around a plausible value, tight precisely *because* the curve is smooth and every resample of a log-linear curve is also log-linear.

An analysis that buckets unmatched symbols into "other" will produce confident output on the wrong alphabet. Our phoneme-class map was keyed on ARPAbet; the sweep emits IPA. Applying one to the other sent 57% of reference tokens, including *every* vowel and /ɪ/, to "other", and still printed per-class slopes, an H2 verdict and a by-L1

table. No error was raised. The guard is to assert coverage rather than infer it from the absence of an exception, which the analysis now does.

A fourth attempt, a TTS-counterfactual substitute for forced alignment, was built and rejected before producing any reported number; it is documented in the repository.

VII. Limitations

The lookahead axis of the reported sweeps is nominal. The sweeps used a per-layer mask, under which the effective future context exceeds L by the amounts in Table I. The measured error rates, and the comparisons between arms at equal L, remain valid as functions of the nominal value, but no budget reported here should be read as an end-to-end algorithmic delay, and the cross-encoder comparison is confounded with tap depth.

Every quality result here is phone-level, not perceptual, and this is a phone-translation probe rather than an end-to-end FAC system: the latency–accuracy relation reported here need not transfer to synthesis, prosody, speaker similarity or perceived accentedness. We do not conduct a perceptual study because the phone-sequence outputs are not valid FAC stimuli: synthesised from bare phone strings, they lack the lexical, prosodic and acoustic structure required to judge naturalness or accentedness. This is a property of the stimuli rather than of the budget. Training is 1200 steps at fixed budget over a frozen encoder, a comparison at matched budget rather than at convergence, though §V-B shows the ranking and exchange rate hold to $5\times$ that budget; the dense sweep has two seeds per arm. Its repeatability threshold now comes from a deliberate four-condition $\times$ four-repeat experiment with a bootstrap interval, but that threshold is global while the per-condition σ it pools varies by a factor of 3.5, so marginal decisions near the threshold should be treated as approximate. t_{buf}'s I/O term is unmeasured, and compute is characterised on Apple Silicon and ARM64 Linux only. The encoder comparison in §V-D compares two pretrained encoder checkpoints. The wav2vec 2.0 result is replicated at a second training seed and re-checked at 6000 steps, while the WavLM curve rests on the seeds of the main sweep; that is enough to show the exchange rate does not transfer, but not to attribute the gap to pretraining scale, objective or architecture; and the layer-selection protocols are not fully symmetric: WavLM's layer 9 was fixed a priori from prior practice, whereas wav2vec 2.0's was selected by a phonetic probe at one lookahead, though re-selection at $L{=}0$ returns the same layer.

VIII. Conclusion

A lookahead budget describes a streaming model only if the output of the complete model depends on no more future input than the budget states. Two conditions for this were found not to hold by default: masking self-attention leaves a pretrained encoder non-causal, and a lookahead mask applied in every layer compounds, so that a nominal 40 ms corresponded here to at least 220 ms. Both are detected by simple input-perturbation tests, and both have direct remedies. For the causal WavLM phone-translation probe under per-layer lookahead, future context reduces PER at an approximately log-linear average rate of ≈ 0.045 per doubling, with no identifiable knee and measurable returns out to a nominal 340 ms, at least 1.5 s of effective context. That rate, however, is measured over one backbone: a second causalised encoder with similar zero-lookahead PER shows only a 3.9% relative endpoint PER reduction from 0 to 640 ms against 63.3% for WavLM, and 1.8% when both are trained to 6000 steps (§V-D). Neither the magnitude nor the detailed shape of the response transfers across the two encoders tested, so future-context budgets should be characterised for the deployed representation rather than inherited from another backbone. Increasing lookahead changes algorithmic latency substantially while only weakly affecting measured per-chunk compute over the tested range, though throughput must still satisfy $t_{\mathrm{cmp}} < c$ (Eq. 8) to run at all.

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X. Compliance with ethical standards

This work used only the publicly released L2-ARCTIC corpus and involved no new participant recruitment, interaction or data collection. No listening study was run and no claim about perceived accentedness is made. The author declares no competing interests.

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